An Explicit Polynomial Net for the Monotone Spherical Chamber

Solution packet for arXiv:1210.0139

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Status. Scoped full result likely valid, subject to human review. At every fixed chordal accuracy, the optimal covering number has polynomial order in the ambient dimension, and a deterministic structure-preserving polynomial-size net is given explicitly. The scope is the source problem's natural fixed-accuracy dimension asymptotic; sharp joint dependence on ε is not claimed.

1 Source question

Fickus, Jasper, Mixon, and Peterson define

$$\mathbb{S}_{\mathrm{nn}}^{M-1} = \{x \in \mathbb{R}^M : 0 \le x_1 \le \cdots \le x_M,\ \|x\|_2 = 1\}$$

and use the chordal distance

$$\rho(x,y) = \sqrt{1 - |\langle x, y\rangle|^2}.$$

Their signed-permutation reduction makes an efficient net for this chamber the computational bottleneck in their NERF estimates. In the conclusions they ask, in effect, how many nonnegative monotone unit vectors are needed to approximate all such vectors in chordal distance [1, p. 16].

Let $\mathcal{N}_M(\varepsilon)$ denote the least cardinality of an ε-net for $\mathbb{S}_{\mathrm{nn}}^{M-1}$ in ρ. The source constructs, for fixed ε, a net of size $\exp(O_\varepsilon((\log M)^2))$. The results below improve the dimension dependence to a power of M, show that a power is necessary at sufficiently fine fixed accuracy, and replace the abstract net by an explicit structure-preserving construction.

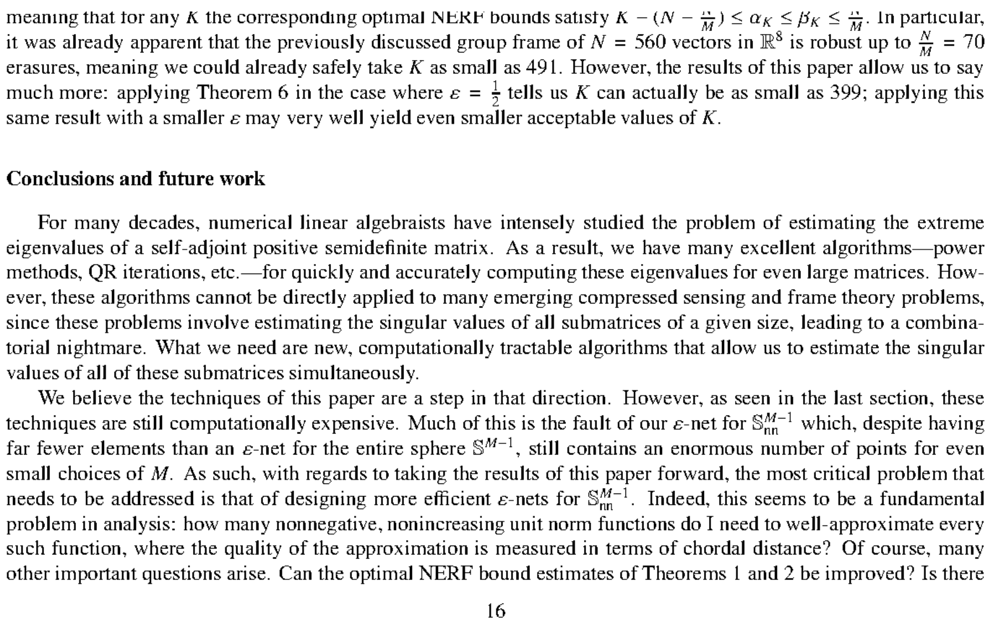
Reversing the order of the coordinates is an isometry. We therefore write

$$T_M = \{x \in \mathbb{R}^M : x_1 \ge \cdots \ge x_M \ge 0,\ \|x\|_2 = 1\}$$

for the coordinate-reversed chamber used in the constructions below.

2 Proof intuition

The upper bound uses monotonicity in the coordinate index rather than quantization in the value axis. Reverse the coordinates and divide them into geometrically growing consecutive blocks. On a block beginning at a, use at most ηa coordinates. Replacing a decreasing vector by its value at the right endpoint of each block loses at most an η fraction of its squared mass. Thus every chamber point is close to a fixed block-constant subspace of dimension $O(\eta^{-1}\log M)$. Taking $\eta \asymp \varepsilon^2$ reduces the problem to a low-dimensional weighted isotonic cone. For the explicit net, round its block coordinates to the integer lattice, project the rounded point back onto that cone, and normalize. Projection cannot increase the rounding error, while a lattice-point volume estimate retains polynomial cardinality.


meaning that for any K the corresponding optimal NERF bounds satisfy $K-(N-\frac{N}{M})\le\alpha_K\le\beta_K\le\frac{N}{M}$. In particular, it was already apparent that the previously discussed group frame of $N=560$ vectors in $\mathbb{R}^8$ is robust up to $\frac{N}{M}=70$ erasures, meaning we could already safely take K as small as 491. However, the results of this paper allow us to say much more: applying Theorem 6 in the case where $\varepsilon=\frac{1}{2}$ tells us K can actually be as small as 399; applying this same result with a smaller ε may very well yield even smaller acceptable values of K.

Conclusions and future work

For many decades, numerical linear algebraists have intensely studied the problem of estimating the extreme eigenvalues of a self-adjoint positive semidefinite matrix. As a result, we have many excellent algorithms—power methods, QR iterations, etc.—for quickly and accurately computing these eigenvalues for even large matrices. However, these algorithms cannot be directly applied to many emerging compressed sensing and frame theory problems, since these problems involve estimating the singular values of all submatrices of a given size, leading to a combinatorial nightmare. What we need are new, computationally tractable algorithms that allow us to estimate the singular values of all of these submatrices simultaneously.

We believe the techniques of this paper are a step in that direction. However, as seen in the last section, these techniques are still computationally expensive. Much of this is the fault of our ε-net for $\mathbb{S}^{M-1}_{\mathrm{nn}}$ which, despite having far fewer elements than an ε-net for the entire sphere $\mathbb{S}^{M-1}$, still contains an enormous number of points for even small choices of M. As such, with regards to taking the results of this paper forward, the most critical problem that needs to be addressed is that of designing more efficient ε-nets for $\mathbb{S}^{M-1}_{\mathrm{nn}}$. Indeed, this seems to be a fundamental problem in analysis: how many nonnegative, nonincreasing unit norm functions do I need to well-approximate every such function, where the quality of the approximation is measured in terms of chordal distance? Of course, many other important questions arise. Can the optimal NERF bound estimates of Theorems 1 and 2 be improved? Is there

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Figure 1: The open-ended covering question in the source paper, page 16.

For the lower bound, reverse the idea. Use $d\asymp\log M$ blocks whose sizes decrease by a factor 16. Give every block nearly equal *energy*, and perturb those energies by balanced signs from a binary code. The factor 16 leaves enough separation between adjacent block heights to preserve monotonicity under every perturbation. Codewords far apart in Hamming distance then give chamber points a fixed positive chordal distance apart.

3 The metric-entropy theorem

Theorem 1. *There are universal constants $c>0$ and $M_0\ge 2$ such that, for every $M\ge M_0$ and $0<\varepsilon\le 1/5$,*

$$M^c\le\mathcal{N}_M(\varepsilon)\le\left(1+\frac{4}{\varepsilon}\right)^{32\varepsilon^{-2}(1+\log M)}. \tag{1}$$

Consequently, for each fixed $0<\varepsilon\le 1/5$,

$$\log\mathcal{N}_M(\varepsilon)=\Theta_\varepsilon(\log M)\qquad(M\to\infty).$$

The upper bound remains valid for every $0<\varepsilon<1$.

The next theorem supplies the constructive part needed for the source's computational formulation.

Theorem 2 (explicit structure-preserving net)**.** *For every $M\ge 2$ and $0<\varepsilon<1$, the geometric partition used below, with*

$$d\le 32\varepsilon^{-2}(1+\log M),$$

determines a finite set $A_{M,\varepsilon} \subseteq T_M$ satisfying

$$|A_{M,\varepsilon}| \le (42/\varepsilon)^d \quad \textit{and} \quad \sup_{x\in T_M} \min_{a\in A_{M,\varepsilon}} \rho(x,a) < \varepsilon.$$

The set can be enumerated deterministically using integer-lattice enumeration, weighted isotonic regression, and normalization, in $O(d(42/\varepsilon)^d)$ arithmetic operations. In particular, for every fixed ε, both its cardinality and its enumeration time are polynomial in M.

For points in either chamber the inner product is nonnegative, and

$$\rho(x,y)^2 = (1-\langle x,y\rangle)(1+\langle x,y\rangle) \le 2(1-\langle x,y\rangle) = \|x-y\|_2^2. \tag{2}$$

Lemma 1 (geometric-block compression)**.** *For every $0 < \eta \le 1$ there is a partition of $\{1,\dots,M\}$ into d consecutive blocks such that*

$$d \le 4\eta^{-1}(1+\log M), \tag{3}$$

and the orthogonal projection P onto the subspace of vectors constant on each block satisfies

$$\|x-Px\|_2^2 \le \eta \qquad (x\in T_M). \tag{4}$$

Moreover, Px is nonnegative and nonincreasing.

Proof. Make every index $a < 2/\eta$ a singleton. Thereafter, whenever the next block starts at a, give it length

$$\ell_a = \min\{\lfloor \eta a\rfloor, M-a+1\}.$$

For a nonterminal nonsingleton block, $\eta a/2 \le \ell_a \le \eta a$. Its successor therefore begins at least at $(1+\eta/2)a$. Since $\log(1+\eta/2) \ge \eta/4$, counting the initial singletons and subsequent geometric growth gives (3).

Fix $x \in T_M$. On a block $B = [a,b]$, define $q_i = x_b$ for $i \in B$. Then q is block-constant, and

$$\sum_{i\in B}(x_i - x_b)^2 \le \sum_{i\in B}(x_i^2 - x_b^2). \tag{5}$$

Put $z_i = x_i^2$. For any $t \ge 0$, the set $\{i : z_i > t\}$ is an initial interval $\{1,\dots,k(t)\}$. In the layer-cake representation of the right side of (5), complete blocks cancel. If $k(t)$ lies in a nonsingleton block $[a,b]$, the uncancelled number of indices is at most $|B| \le \eta a \le \eta k(t)$; it is zero in an initial singleton. Hence

$$\|x-q\|_2^2 \le \int_0^\infty \eta k(t)\,dt = \eta\sum_{i=1}^M z_i = \eta.$$

Orthogonal projection is at least as accurate as q, proving (4). Finally, Px consists of the consecutive block averages of x, so it remains nonnegative and nonincreasing. $\square$

Proof of Theorem 2. Use Lemma 1 with $\eta = \varepsilon^2/8$. Write its blocks as $B_1,\dots,B_d$, put $w_j = |B_j|$, and define the isometry

$$J : \mathbb{R}^d \longrightarrow V, \qquad (Jz)_i = \frac{z_j}{\sqrt{w_j}} \quad (i \in B_j).$$

The block-constant chamber corresponds under J to the closed convex cone

$$K = \left\{ z \in \mathbb{R}^d : \frac{z_1}{\sqrt{w_1}} \ge \cdots \ge \frac{z_d}{\sqrt{w_d}} \ge 0 \right\}.$$

Let Π_K be Euclidean metric projection onto K, and set

$$Q = \left\lceil \frac{8\sqrt{d}}{\varepsilon} \right\rceil, \qquad \Lambda_Q = \left\{ k \in \mathbb{Z}^d : \|k\|_2 \le Q + \frac{\sqrt{d}}{2} \right\}.$$

The promised net is

$$A_{M,\varepsilon} = \left\{ J \frac{\Pi_K k}{\|\Pi_K k\|_2} : k \in \Lambda_Q,\ \Pi_K k \ne 0 \right\}. \tag{6}$$

Every member is nonnegative, nonincreasing, block-constant, and unit norm.

To verify covering, fix $x \in T_M$ and let $u = Px/\|Px\|_2$. If $e = \|x - Px\|_2$, orthogonality gives $\|Px\|_2 = \sqrt{1-e^2}$ and therefore

$$\|x-u\|_2^2 = 2(1 - \|Px\|_2) \le 2e^2 \le 2\eta = \varepsilon^2/4.$$

Put $z = J^{-1}u \in K \cap S^{d-1}$ and obtain $k \in \mathbb{Z}^d$ by rounding every coordinate of Qz to the nearest integer. Then $k \in \Lambda_Q$ and

$$\|k/Q - z\|_2 \le \frac{\sqrt{d}}{2Q} \le \frac{\varepsilon}{16}.$$

Metric projection fixes z and is nonexpansive. Hence, for $v = \Pi_K(k/Q) = \Pi_K(k)/Q$,

$$\|v - z\|_2 \le \varepsilon/16, \qquad \left\| \frac{v}{\|v\|_2} - z \right\|_2 \le 2\|v-z\|_2 \le \varepsilon/8.$$

Thus the corresponding point of (6) is within $5\varepsilon/8$ of x in Euclidean distance, and therefore within ε in chordal distance by (2).

It remains to count the candidates. The disjoint unit cubes centered at points of Λ_Q lie in the Euclidean ball of radius $Q + \sqrt{d}$. Writing v_d for the volume of the d-dimensional unit ball and using $v_d \le (2\pi e/d)^{d/2}$ gives

$$|\Lambda_Q| \le v_d(Q + \sqrt{d})^d \le \left(\frac{10\sqrt{2\pi e}}{\varepsilon} \right)^d < \left(\frac{42}{\varepsilon} \right)^d.$$

Finally, $\Pi_K k$ is exactly a weighted isotonic regression problem after writing $z_j = \sqrt{w_j} h_j$; the pool-adjacent-violators algorithm computes it in $O(d)$ operations. Recursive enumeration of the integer points in Λ_Q therefore gives the stated output-sensitive running time. □

Proof of the upper bound in Theorem 1. Take $\eta = \varepsilon^2/8$ and the subspace V from Lemma 1. If $e = \|x - Px\|_2$, orthogonality gives $\|Px\|_2 = \sqrt{1-e^2}$. Thus, for $u = Px/\|Px\|_2 \in T_M \cap V$,

$$\|x-u\|_2^2 = 2(1 - \|Px\|_2) \le 2e^2 \le 2\eta = \varepsilon^2/4. \tag{7}$$

Take a maximal Euclidean $\varepsilon/2$-separated subset A of $T_M \cap V$. It is an $\varepsilon/2$-net of that set. The usual disjoint-ball argument in the d-dimensional space V gives

$$|A| \le (1 + 4/\varepsilon)^d.$$

Equation (7) and the triangle inequality show that A is an Euclidean ε-net of T_M, and (2) makes it a chordal ε-net as well. Since Lemma 1 gives $d \le 32\varepsilon^{-2}(1 + \log M)$, the upper estimate follows. □

Proof of the lower bound in Theorem 1. Let d be the largest even integer for which

$$m_d := 1 + 16 + \cdots + 16^{d-1} = \frac{16^d - 1}{15} \le M.$$

Thus $d \ge c_1 \log M$ for all sufficiently large M. After $M - m_d$ initial zero coordinates, split the remaining coordinates into consecutive blocks $B_1, \ldots, B_d$ of sizes $|B_j| = 16^{d-j}$.

We use a family $\mathcal{C} \subseteq \{-1,1\}^d$ of balanced words ($\sum_j \sigma_j = 0$) with pairwise Hamming distance at least $d/4$ and $|\mathcal{C}| \ge \exp(c_2 d)$. For completeness, such a family follows by a greedy selection from the $\binom{d}{d/2}$ balanced words. A Hamming ball of radius less than $d/4$ contains at most

$$\sum_{r \le d/8} \binom{d/2}{r}^2 \le (d+1)2^{dH_2(1/4)}$$

balanced words, while $\binom{d}{d/2} \ge 2^d/(d+1)$ and $H_2(1/4) < 1$.

For $\sigma \in \mathcal{C}$, define x^σ to be constant on B_j with value

$$x_i^\sigma = \frac{1 + \sigma_j/2}{\sqrt{(5/4)d\,|B_j|}} \qquad (i \in B_j). \tag{8}$$

Balancedness gives $\|x^\sigma\|_2 = 1$. Adjacent block heights obey

$$\frac{x^\sigma|_{B_{j+1}}}{x^\sigma|_{B_j}} = 4\frac{1 + \sigma_{j+1}/2}{1 + \sigma_j/2} \ge \frac{4}{3},$$

so every x^σ belongs to the nonnegative nondecreasing chamber.

If h is the Hamming distance between σ and τ, direct calculation using balancedness gives

$$\langle x^\sigma, x^\tau \rangle = 1 - \frac{2(1/2)^2}{1 + (1/2)^2}\frac{h}{d} = 1 - \frac{2}{5}\frac{h}{d}.$$

For $h \ge d/4$, this is at most $9/10$, and hence

$$\rho(x^\sigma, x^\tau) \ge \sqrt{1 - (9/10)^2} = \frac{\sqrt{19}}{10} > \frac{2}{5}.$$

When $\varepsilon \le 1/5$, an ε-ball contains at most one member of this packing. Therefore

$$\mathcal{N}_M(\varepsilon) \ge |\mathcal{C}| \ge \exp(c_2 d) \ge M^c,$$

after decreasing the universal constant $c > 0$ and increasing M_0. □

4 Verification, scope, and novelty

The argument is elementary. The accompanying script `code/verify_entropy_bounds.py` checks the geometric-block estimate on random and adversarial monotone vectors, verifies the dimension count, and enumerates finite balanced-code packings to check normalization, monotonicity, and the chordal separation constant. These computations are sanity checks, not ingredients of the proof. It passed with 300 random trials at $(M, \varepsilon, d) = (4096, 0.2, 8)$ and 200 trials at $(7777, 0.125, 10)$. The commands were

```
python code/verify_entropy_bounds.py --dimension 4096 --epsilon 0.2 \
  --trials 300 --packing-blocks 8
python code/verify_entropy_bounds.py --dimension 7777 --epsilon 0.125 \
  --trials 200 --packing-blocks 10
```

In the first run, the minimum enumerated packing distance was exactly $\sqrt{19}/10$, the constant used in the proof. The separate script `code/verify_explicit_net.py` checks the rounding, weighted isotonic projection, normalization, chamber preservation, and the $5\varepsilon/8$ covering estimate used in Theorem 2. It passed 303 test vectors at $(M, \varepsilon) = (4096, 0.2)$ and 203 test vectors at $(7777, 0.125)$ with the commands

```
python code/verify_explicit_net.py --dimension 4096 --epsilon 0.2 --trials 300
python code/verify_explicit_net.py --dimension 7777 --epsilon 0.125 --trials 200
```

These computations test the implementation and constants; the projection and covering arguments above, rather than the finite tests, constitute the proof.

Theorems 1 and 2 settle the source problem's fixed-accuracy dimension-asymptotic formulation: polynomially many points are necessary, polynomially many suffice, and such a net is explicitly enumerable inside the chamber. This is the scope of the full-result classification. The argument does not determine the sharp dependence of the polynomial exponent on ε, so no sharp joint (M, ε) entropy law is claimed.

A bounded novelty search used the run indexes, arXiv:1210.0139, the exact source phrases "nonnegative, nonincreasing unit norm" and "more efficient ε-nets," and combinations of "metric entropy," "monotone cone," "isotonic cone," "chordal distance," "Hellinger entropy," and "monotone probability vectors." It also checked the adjacent bounded monotone-function entropy literature. No later paper directly settling this chamber covering question was found. This is a bounded search, not an exhaustive priority claim.

Human-review recommendation. Check the layer-cake cancellation in Lemma 1, the normalization estimate (7), the lattice count and isotonic projection in Theorem 2, and the constant-weight-code packing.

References

[1] M. Fickus, J. Jasper, D. G. Mixon, and J. Peterson, *Group-theoretic constructions of erasure-robust frames*, Linear Algebra Appl. 479 (2015), 131–154; arXiv:1210.0139.